

Hardy-Weinberg Equilibrium and Population Genetics Practice Problems

1. Sickle Cell Trait and Malaria Resistance (Autosomal)

In a West African population studied in the 1950s, the frequency of the sickle cell allele (β^S) at the β -globin locus was found to be 0.12. Assuming the population is in Hardy-Weinberg equilibrium (HWE) for this locus:

- What are the expected frequencies of the three genotypes (β^A/β^A , β^A/β^S , β^S/β^S)?
- If the population size is 10,000, how many individuals are expected to be heterozygous?
- A follow-up study showed that the frequency of β^S was significantly lower in children under 5 than in adults. What is one likely violation of HWE that this observation suggests?

2. The CCR5- Δ 32 Deletion and HIV Resistance

The CCR5- Δ 32 allele is a 32-base-pair deletion that confers resistance to HIV-1 infection in homozygotes. In a study of a European-derived population in the United States, 1,000 individuals were genotyped: 720 were wild-type homozygotes (+/+), 260 were heterozygotes (+/ Δ 32), and 20 were Δ 32/ Δ 32 homozygotes.

- Calculate the observed allele frequencies of the wild-type (+) and Δ 32 alleles.
- Calculate the expected genotype counts under HWE. Is this population in HWE for this locus? Perform a chi-square test (critical value for 1 df = 3.84).
- If this locus were in HWE, what would you expect the frequency of Δ 32/ Δ 32 homozygotes to be? How does this compare to the observed number, and what might this suggest about the selective advantage of the Δ 32 allele in the context of HIV?

3. X-Linked Locus: Color Blindness in Humans

Red-green color blindness is an X-linked recessive trait. In a large, randomly mating human population, the frequency of color-blind males is 8% (0.08).

- What is the frequency of the color-blindness allele (X^c) in the population?
- What is the expected frequency of color-blind females in this population under HWE?
- If the population is in HWE for this X-linked locus, what is the expected ratio of color-blind males to color-blind females? How does this differ from the ratio for a rare autosomal recessive disorder?

4. X-Linked Locus: Calico Cats

The orange/black coat color in cats is controlled by an X-linked locus with two alleles: O (orange) and o (black). Heterozygous females (O/o) are calico (a mix of orange and black patches due to X-inactivation). In a population of feral cats in a coastal town, you observe the following phenotypes: 45% of males are orange, 55% of males are black. Among females, 36% are orange, 48% are calico, and 16% are black.

- Calculate the allele frequency of O from the male data and from the female data separately. Are they similar? What does this suggest about whether the population is in HWE for this X-linked locus?

5. HWE as a Null Model in Conservation

A population of endangered Florida panthers (*Puma concolor coryi*) was genotyped at a microsatellite locus. The observed genotype counts were: AA = 42, Aa = 6, aa = 2 (total N = 50).

- Calculate the observed allele frequencies.
- Calculate the expected genotype counts under HWE.
- The observed number of heterozygotes is much lower than expected under HWE. What does this suggest about mating patterns in this small, isolated population?

6. Overdominance: Sickle Cell Trait and Malaria

In a particular West African population, the relative fitnesses of the β -globin genotypes are:

- β^A/β^A (normal): $w_{AA} = 0.8$ (due to malaria susceptibility)
- β^A/β^S (heterozygote): $w_{AS} = 1.0$ (resistant to malaria)
- β^S/β^S (sickle cell anemia): $w_{SS} = 0.2$ (due to severe anemia)

- Calculate the equilibrium frequency of the β^S allele ($\hat{q}$) under this overdominant selection model.
- If malaria eradication programs were completely successful in this region, what would happen to the relative fitness of the heterozygote? Predict how the frequency of β^S would change over generations under the new fitness regime.

7. Overdominance in Agriculture: The Case of Cassava Mosaic Disease

Cassava farmers in East Africa cultivate both resistant and susceptible varieties. A locus confers resistance to cassava mosaic virus, with genotypes: RR (highly resistant, but low yield), Rr (resistant and high yield), and rr (susceptible, low yield). The relative fitnesses are $w_{RR} = 0.7$, $w_{Rr} = 1.0$, $w_{rr} = 0.4$.

- What is the equilibrium frequency of the resistance allele (R)?
- A new, more aggressive strain of the virus emerges that can overcome the Rr heterozygote resistance, reducing its fitness to 0.6. What is the new equilibrium? What does this imply about the sustainability of using this single resistance gene in agriculture?

8. Underdominance: Chromosomal Rearrangements in *Anopheles gambiae*

In the mosquito *Anopheles gambiae*, a major malaria vector, a chromosomal inversion (2La) is associated with aridity tolerance. Suppose the fitnesses for genotypes are: AA (standard) = 1.0, Aa (heterozygote) = 0.7, aa (inversion homozygote) = 1.0. This is a classic underdominant (heterozygote disadvantage) scenario.

- If the frequency of the inversion (a) is 0.2, will it increase or decrease in the next generation? (Calculate the new frequency after one generation of selection).
- If the frequency of a is 0.5, will the population move towards fixation of A or fixation of a ? Explain why.

9. Underdominance and Hybrid Zones

Two subspecies of mice (*Mus musculus domesticus* and *M. m. musculus*) meet in a hybrid zone in Europe. They differ in a Robertsonian fusion/fission that causes heterozygote infertility. The relative fitnesses are: $w_{AA} = 1.0$, $w_{Aa} = 0.6$, $w_{aa} = 1.0$, where A is the domesticus allele and a is the musculus allele.

- Calculate the unstable equilibrium frequency of the a allele.
- Describe what would happen to a population that initially has a frequency of $a = 0.3$ versus a population with $a = 0.7$. Which allele will become fixed in each case?

10. Selection and Purging of a Lethal Allele

A lethal recessive allele (d) exists at a locus. The fitnesses are: $DD = 1.0$, $Dd = 1.0$, $dd = 0.0$. The current frequency of the d allele is 0.1.

- Assuming Hardy-Weinberg proportions before selection, calculate the genotype frequencies in the population.
- Apply selection and calculate the genotype frequencies after selection (before normalization).
- Calculate the mean fitness of the population and normalize the genotype frequencies.
- Using the normalized genotype frequencies, calculate the frequency of the d allele in the next generation.
- If the population continues to reproduce randomly, will the d allele eventually be completely eliminated? Why or why not?

11. Unidirectional Mutation: Congenital Hypertrichosis

Congenital hypertrichosis (excessive hair growth) is a rare dominant condition. Suppose the rate of mutation from the wild-type allele (h) to the dominant disease allele (H) is 1×10^{-5} per generation. The fitness of individuals with the disease (HH and Hh) is 0.7 relative to wild-type homozygotes (hh).

- What is the equilibrium frequency of the dominant disease allele (H) under mutation-selection balance?
- Why is the equilibrium frequency for a dominant deleterious allele much lower than that for a recessive deleterious allele with the same mutation rate and selection coefficient?

12. Recessive Lethal: Tay-Sachs Disease

Tay-Sachs disease is a fatal autosomal recessive disorder caused by mutations in the HEXA gene. In the general population, the incidence is about 1 in 320,000 births. In Ashkenazi Jewish populations, the incidence is historically about 1 in 3,600 births.

- Estimate the frequency of the Tay-Sachs allele (t) in the general population and in the Ashkenazi Jewish population, assuming mutation-selection balance with $s = 1$ (lethal).
- What is the estimated mutation rate from the wild-type to the Tay-Sachs allele in the Ashkenazi Jewish population, based on these data?
- The mutation rate estimated from the Ashkenazi Jewish population is much higher than the actual mutation rate. What evolutionary force (other than mutation) likely explains the high frequency of this allele in this population?

13. Bidirectional Mutation and Equilibrium

At a certain locus, the forward mutation rate from allele A to a is 1×10^{-5} and the backward mutation rate from a to A is 5×10^{-6} . Assume no selection.

- What is the equilibrium frequency of the a allele ($\hat{q}$) under bidirectional mutation alone?
- If you start with a population fixed for A , will the frequency of a ever reach 0.5? Explain why or why not.

14. Mutation-Selection Balance in Antibiotic Resistance

In the absence of antibiotics, a mutation conferring rifampicin resistance in *Mycobacterium tuberculosis* carries a fitness cost, $s = 0.2$. Assume that resistance is **recessive** (i.e., only resistant homozygotes suffer the fitness cost in the absence of antibiotics). The mutation rate from sensitive to resistant is 1×10^{-8} .

- What is the equilibrium frequency of resistant bacteria in a population not exposed to rifampicin?
- When a patient is treated with rifampicin, the selection coefficient reverses ($s < 0$ for sensitive, $s > 0$ for resistant). What happens to the equilibrium frequency of resistant bacteria during treatment? Explain why stopping treatment prematurely can lead to a relapse with a resistant infection.

15. Mutation-Selection Balance in a Conservation Context

A small, isolated population of mountain gorillas has a population size of 500. A new recessive deleterious mutation arises at a rate of 1×10^{-6} per generation. The mutation causes juvenile sterility with a selection coefficient $s = 0.5$.

- Calculate the expected equilibrium frequency of this deleterious allele (q) under mutation-selection balance.
- If the population size remains small, would you expect the actual frequency of this allele to be exactly at the equilibrium you calculated, or could it deviate? Explain your reasoning.

16. Gene Flow and the Sickle Cell Allele

Consider an "island" population of African descent living in the United States (where malaria is absent) and a "continent" population in West Africa (where malaria is present). At the β -globin locus, the frequency of the β^S allele on the continent is $q_m = 0.12$. On the island, it is $q_0 = 0.06$.

On the island, assume the following fitnesses: $w_{AA} = 1.0$, $w_{AS} = 1.0$, $w_{SS} = 0.2$.

Assume that migration occurs **before selection** each generation, and that a fraction m of the island population consists of migrants from the continent.

- Write an expression for the allele frequency after migration in terms of q_0 , q_m , and m .
- Write an expression for the allele frequency after selection in terms of genotype fitnesses.
- Assuming the population is at equilibrium, estimate the migration rate m .

17. Gene Flow and Pesticide Resistance in Crop Pests

A population of cotton bollworms (*Helicoverpa armigera*) in a cotton-growing region (island) is exposed to the pesticide pyrethroid. A resistance allele (R) has a fitness advantage in the presence of the pesticide. However, a large refuge population of susceptible individuals (continent) exists on unsprayed wild host plants, with $q_m = 0$ (no resistance). In the cotton field, the relative fitnesses are: $RR = 1.0$, $Rr = 1.0$, $rr = 0.6$. The frequency of R in the cotton field at the start of the season is 0.8.

- a) If the migration rate from the refuge to the cotton field is 0.1 per generation, what will be the frequency of R in the cotton field in the next generation, before selection acts again?
- b) This is a form of "refuge strategy" in pest management. Explain how the refuge (continent) helps to delay the evolution of pesticide resistance, even though the resistance allele is advantageous in the cotton field.

18. Counteracting Gene Flow: Conservation of an Endangered Plant

An endangered plant species, *Brighamia insignis* (Hawaiian lobelioid), exists in a single wild population (island) of 50 individuals. A related, weedy species (*Brighamia rockii*) grows nearby and hybridizes, introducing a maladaptive allele (A_2) at a locus affecting drought tolerance. The weedy species (continent) is fixed for A_2 . In the endangered population, the native allele A_1 has a fitness advantage, with $w_{A_1A_1} = 1.0$, $w_{A_1A_2} = 0.8$, $w_{A_2A_2} = 0.6$. The current frequency of A_2 in the endangered population is 0.1. The migration rate from the weedy species is 0.05 per generation.

Assume that migration occurs **before selection** each generation, and that Hardy-Weinberg proportions apply before selection.

- a) What is the expected frequency of A_2 after migration?
- b) Using Hardy-Weinberg proportions and the given fitnesses, calculate the frequency of A_2 after selection in the next generation.
- c) What is the equilibrium frequency of A_2 in the endangered population under this migration-selection balance?
- d) What conservation action could be taken to reduce the introduction of the maladaptive A_2 allele into the endangered population?

19. Island-Continent Model for an X-Linked Locus

A rare X-linked recessive allele (c) causing a mild disorder has a frequency of 0.01 on the mainland (continent). On an isolated island, the frequency is 0.00. Every generation, 2% of the island's male population are immigrants from the mainland. Assume no selection and random mating on the island. What will be the frequency of the c allele in island males and in island females after one generation of migration? (Hint: Treat male and female frequencies separately. Males inherit their X from their mother, while females inherit one X from their father and one from their mother.)

20. Asymmetric Gene Flow in a Hybrid Zone

Two closely related species of warblers, *Setophaga auduboni* (Audubon's warbler, A) and *S. coronata* (myrtle warbler, a), hybridize in a narrow zone in western Canada. A plumage locus shows strong selection against heterozygotes (underdominance) in the hybrid zone. However, there is also asymmetric migration: more A individuals move into the a territory than vice versa, due to habitat preference. The selection coefficients are: $w_{AA} = 1$, $w_{Aa} = 0.7$, $w_{aa} = 1$. The migration rate from A to a is $m_A = 0.05$, and from a to A is $m_a = 0.01$.

- a) Which species' range is likely to expand due to this asymmetric gene flow?
- b) How does the interaction between migration and underdominant selection influence the stability of the hybrid zone?